

CAMELS-DE-1h Inverse LSTM — Results Summary

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Experiment: LSTM_CAMELS_DE_1h_benchmark
Task: Inverse precipitation-from-discharge (hourly)
Model: CudaLSTM, hidden_size=128, seq_length=96
Training period: 2004-01-01 to 2019-12-31
Testing period: 2020-01-01 to 2024-12-31
Epochs trained: 20 (checkpoint used for testing: epoch 20)
Catchments evaluated: 216 of 221

Testing NSE (n=216 gauges):

Mean: 0.325
Median: 0.327
Std: 0.057
Min: 0.164
Max: 0.496
Gauges with NSE < 0: 0
Gauges with NSE > 0.5: 0

Testing RMSE (n=216 gauges):

Mean: 0.400 mm/hr
Median: 0.394 mm/hr

Note: this is the inverse task (predicting precipitation from discharge), which is substantially harder than the standard forward rainfall-runoff direction. The CAMELS-DE-1h benchmark LSTM (forward task) achieves a median NSE of 0.82; this is not directly comparable.

Figure 1: Testing NSE across catchments

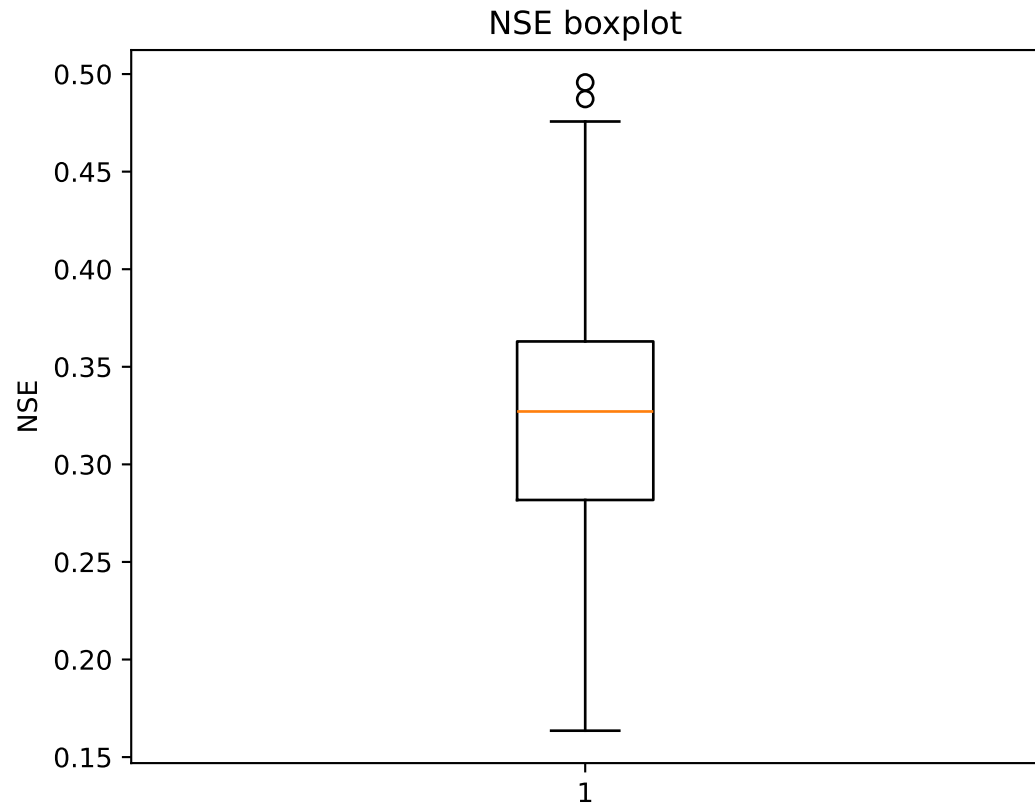
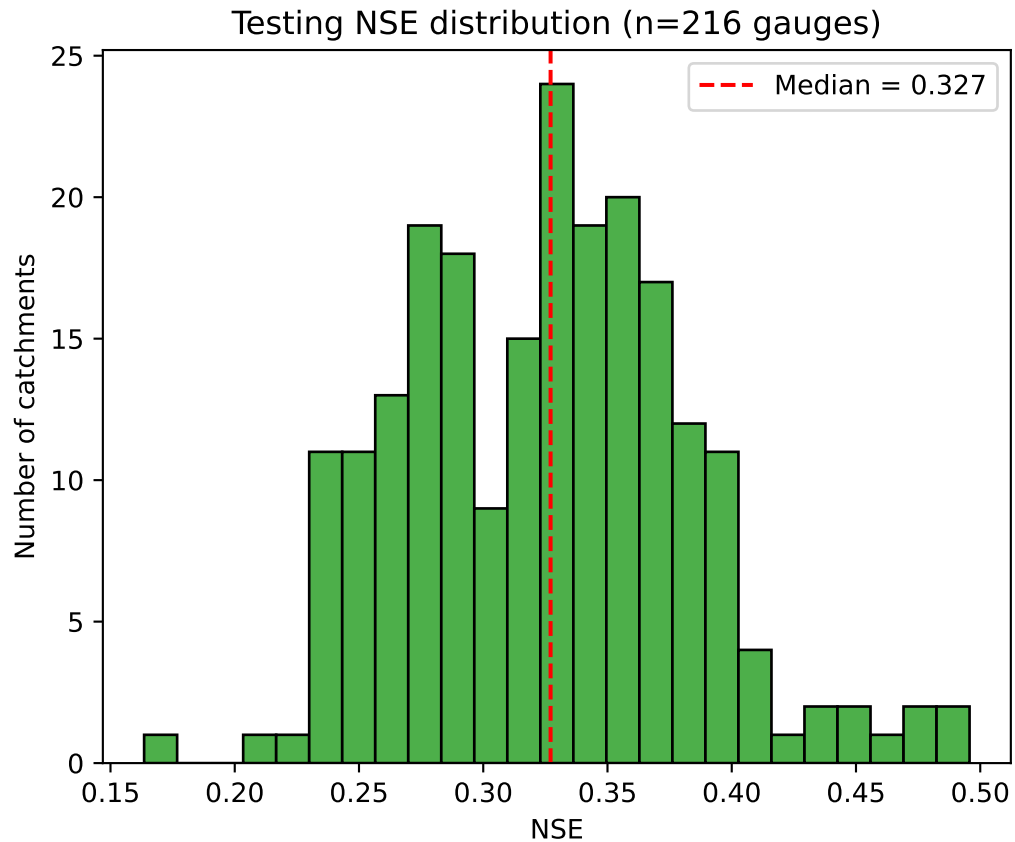


Figure 2: Training loss vs. validation NSE

